Digital Twin Challenge 2026 · Happiest Health

GlucoTwin

A personal digital twin that warns of post-meal glucose spikes before they happen, and lets a doctor test what would prevent them.

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The problem

India has about **101 million** people with diabetes and **136 million** with prediabetes.

ICMR-INDIAB national study, *Lancet Diabetes & Endocrinology*, 2023

Much of the damage happens in the hours after meals. Rice- and roti-heavy meals drive sharp spikes above 180 mg/dL, and repeated spikes drive complications.

Doctors see these spikes after the fact, in an HbA1c report or a CGM download, when it is too late to act on that meal.

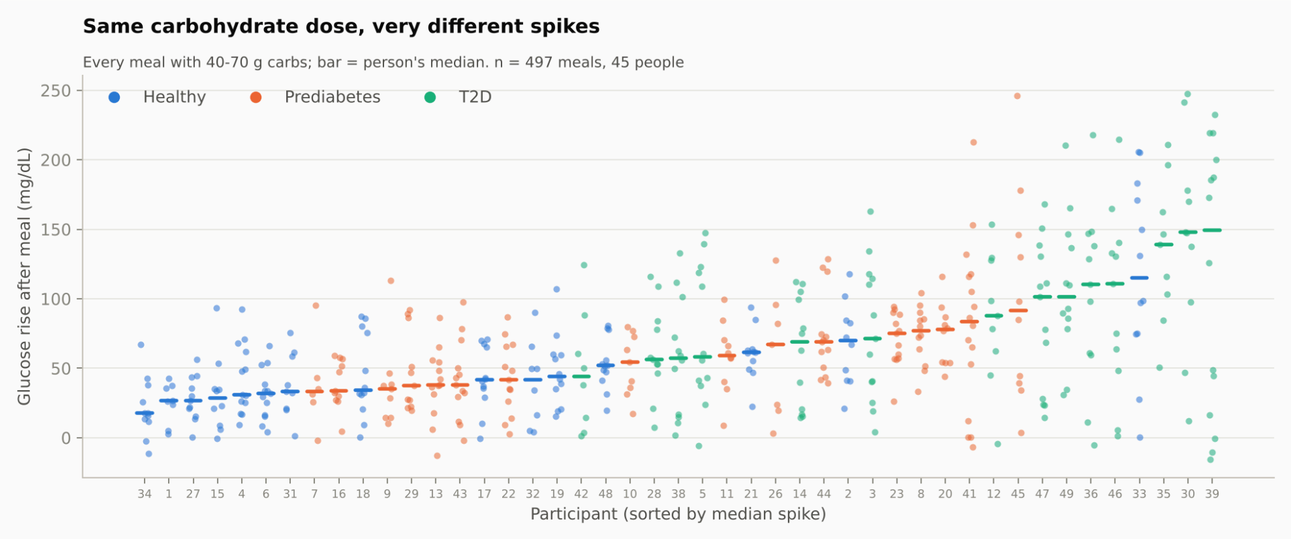
Two kinds of data, kept apart

Health records: HbA1c, fasting insulin, BMI, diagnoses, medication, family history, genetics.

Wearables: glucose every 5 minutes, heart rate, activity, meals.

Our question: can a personal model of each patient, fed by both, predict a spike before it happens?

Why it has to be personal



Every 40-70 g carb meal in CGMacros, one column per person

2%
of a spike's size explained by carbs alone

27%
with the full meal and fasting labs

36%
once the model knows who the person is

Out-of-sample R^2, 5-fold CV, 1,419 meals. Same carb dose: rises from 12 to 132 mg/dL (10th to 90th percentile).

GlucoTwin: a virtual patient for each person

1 · The twin

A physiological model of glucose and insulin, calibrated to one patient. It starts from their health record on day 0 and sharpens with every day of wearable data.

2 · Early warning

Every 15 minutes: glucose forecast 30, 60 and 120 minutes ahead, and the probability of crossing 180 mg/dL in the next 2 hours, with the reasons.

3 · What-if

The doctor tests a plan before the patient eats: rajma chawal now, half a portion, or the same meal and a 20-minute walk. The twin simulates each one.

Decision support only. Every alert is for a clinician to review.

Data: real where it exists, synthetic only where it must be

Stream	Source	
Labs, demographics, BMI	CGMacros bio.csv	real
Dexcom G6 + Libre Pro CGM	CGMacros	real
Fitbit heart rate, METs	CGMacros	real
Meals with macros	CGMacros	real
Diagnoses, medication, family history, TCF7L2 genotype, HbA1c history	Generated from each person's real labs	synthetic

45 adults (15 healthy, 16 prediabetes, 14 T2D by measured HbA1c), 1,684 meals, about 10 days each. Synthetic EHR is exported as FHIR R4, the format of India's ABDM.

Data checks we made

HbA1c is labelled mmol/mol in the dictionary but its values are %. Fixed.

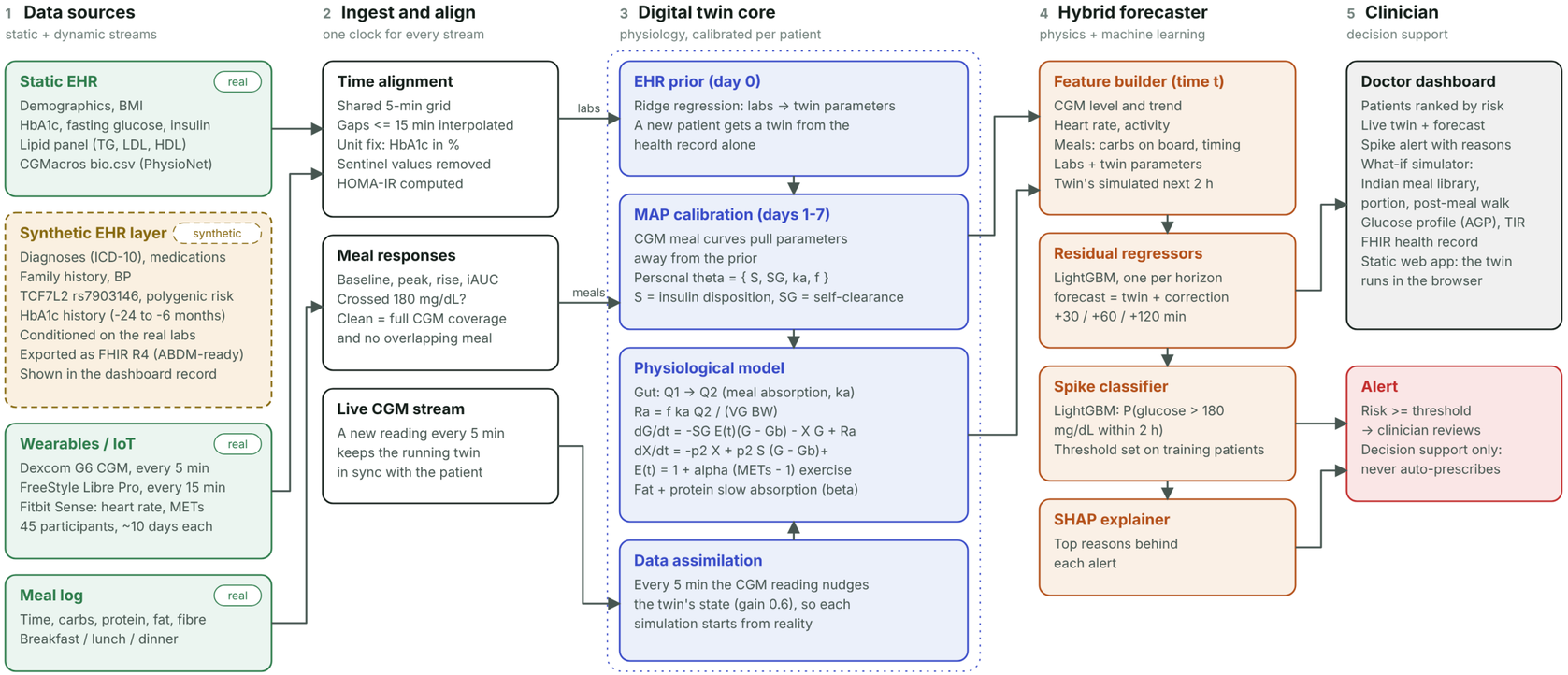
The two CGMs disagree by 25% on average. Against lab fasting glucose, Dexcom was off by +12 mg/dL and Libre by −27, so Dexcom is the primary sensor.

Documented error codes (LDL = 800, VLDL = 400) removed; Fitbit METs rescaled.

Architecture

GlucoTwin system architecture

A per-patient digital twin for Type 2 Diabetes that fuses EHR and wearable data to forecast glucose and warn of post-meal spikes up to 2 hours ahead



Evaluation protocol

Unseen patients
5-fold split by patient

Unseen days
only each patient's held-out last 3 days are scored

Baselines
no change, twin only, ML only

Clinical accuracy
Clarke Error Grid zones A-E

Alerts
detection rate, lead time, false alarms per day

Ablations
what each data stream adds, synthetic fields included

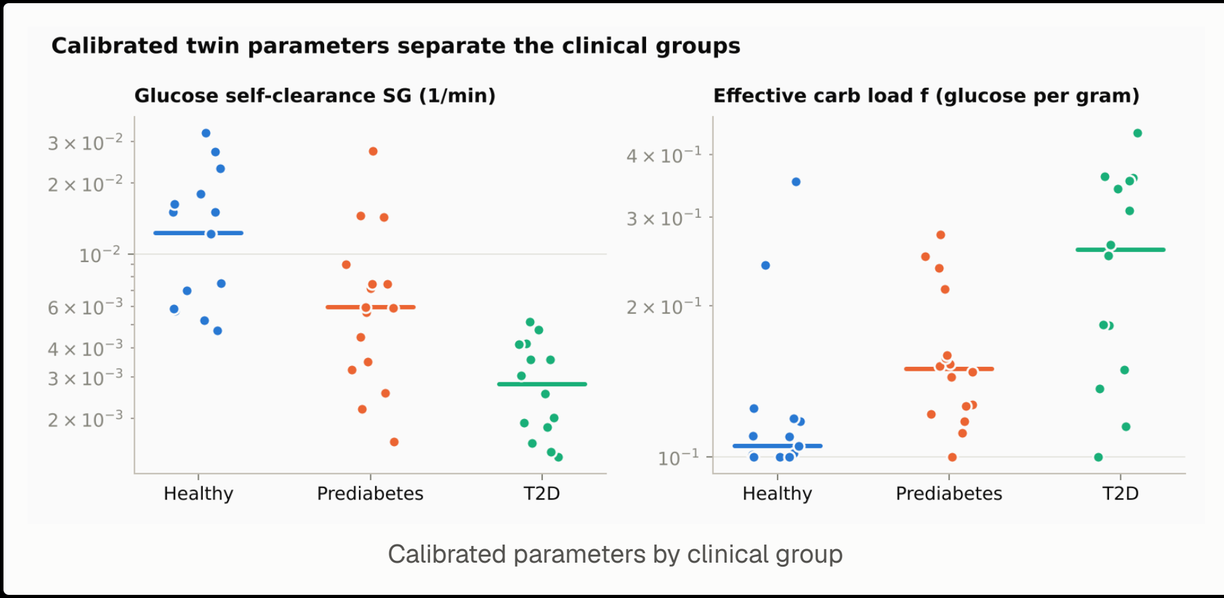
The twin: physiology with four personal numbers

```
dG/dt = −SG·E(t)·(G − Gb) − X·G + Ra(t)
dX/dt = −p2·X + p2·S·max(G − Gb, 0)
Ra(t) = f·ka·Q2 / (VG·BW)     gut: Q1 → Q2
E(t)  = 1 + α·(METs − 1)      exercise
```

SG glucose clearance · **f** effective carb load · **ka** absorption · **S** insulin-driven clearance

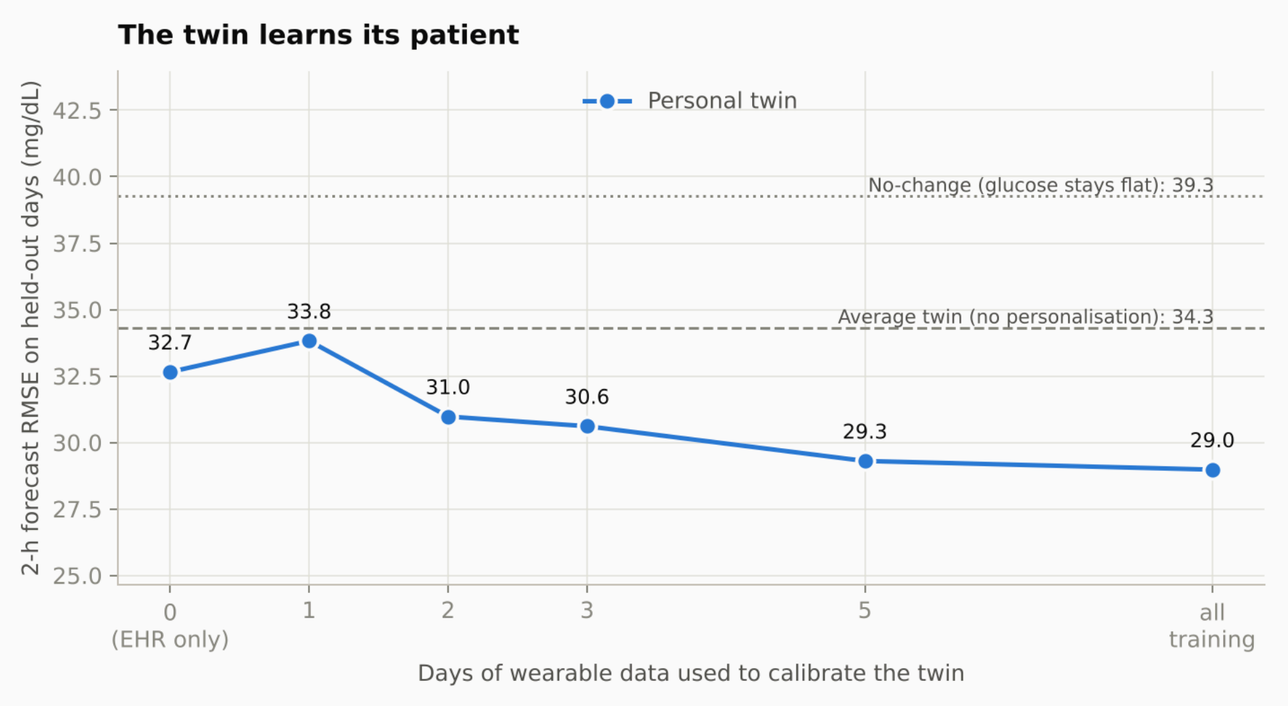
The calibrated twin separates the groups: **SG** falls about 4× from healthy to T2D ($p < 0.0001$) and **f** rises about 2.5× ($p = 0.00032$).

Reduced Bergman minimal model (1979) with a two-compartment gut. CGMacros has no plasma insulin, so S is not identifiable from CGM alone ($p = 0.39$), and we do not interpret it.



Calibrated parameters by clinical group

EHR starts the twin. Wearable data makes it personal.



2-hour forecast error on each patient's held-out last 3 days

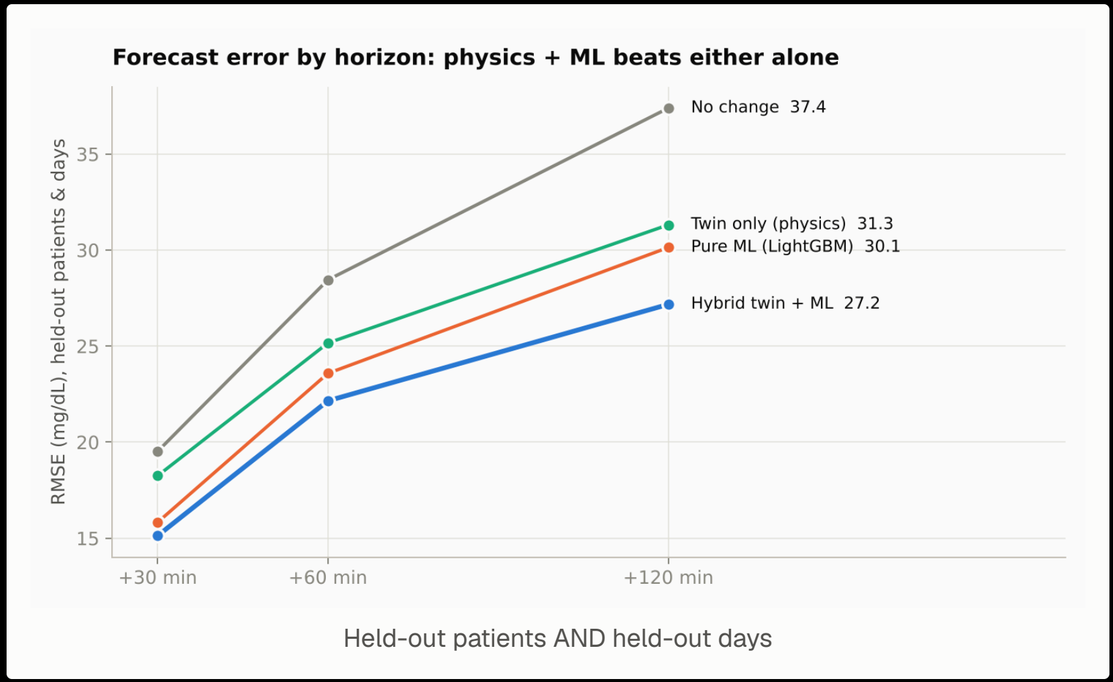
32.7
mg/dL error, day 0: twin built from the EHR alone

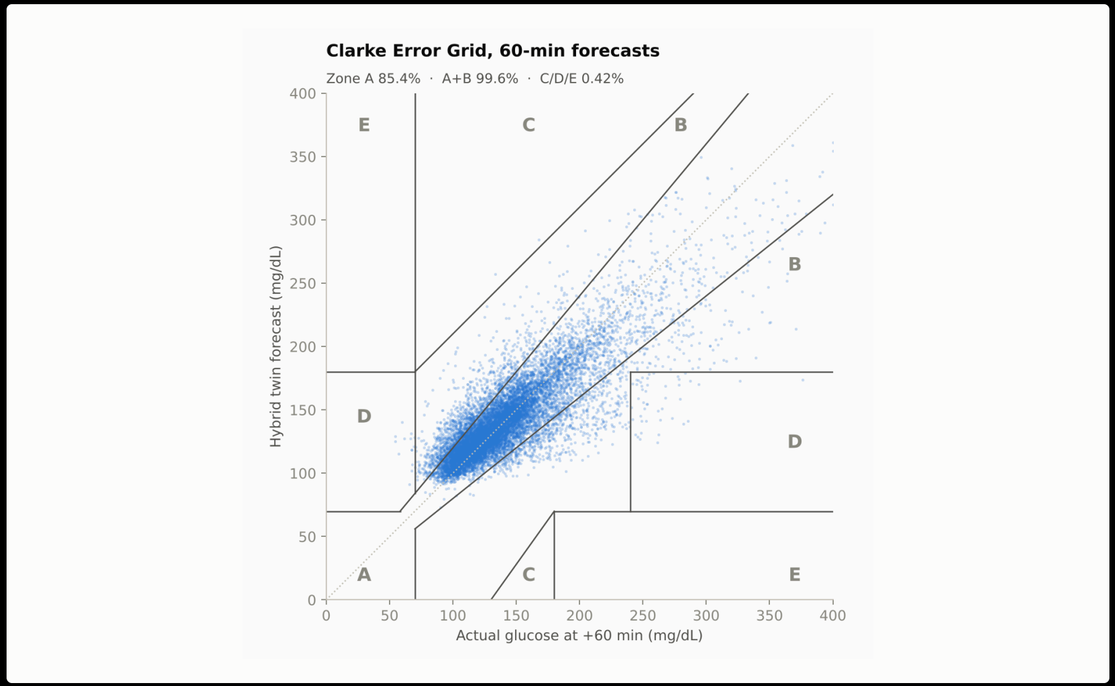
29.0
mg/dL after all training days of CGM

34.3
mg/dL for one average twin for everyone

Calibration is a MAP fit: CGM meal curves pull the parameters away from a prior predicted from labs. Few days, the record dominates; more days, the patient's own data does.

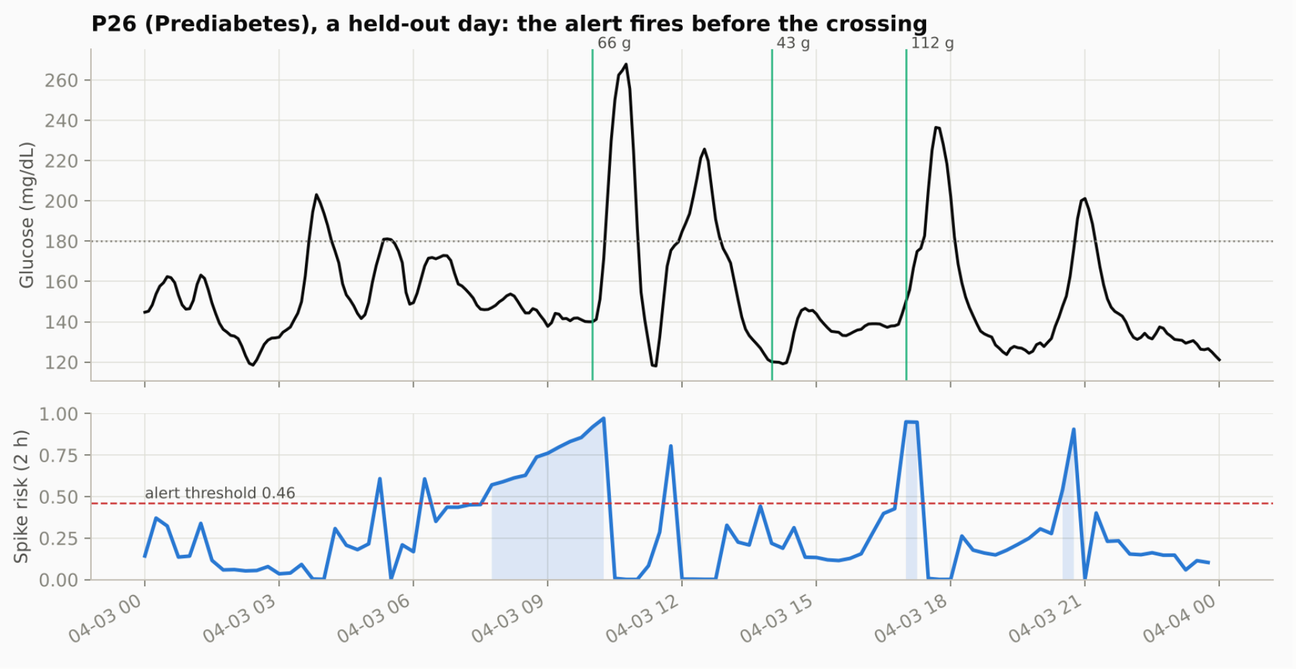
Hybrid forecaster: physics + machine learning





LightGBM learns the twin's error: **forecast = twin + correction**. 60-min error **22.1 mg/dL** · Clarke zone A **85.4%** · zones A+B **99.6%**

Spike alerts a doctor can act on



A held-out day the model never saw

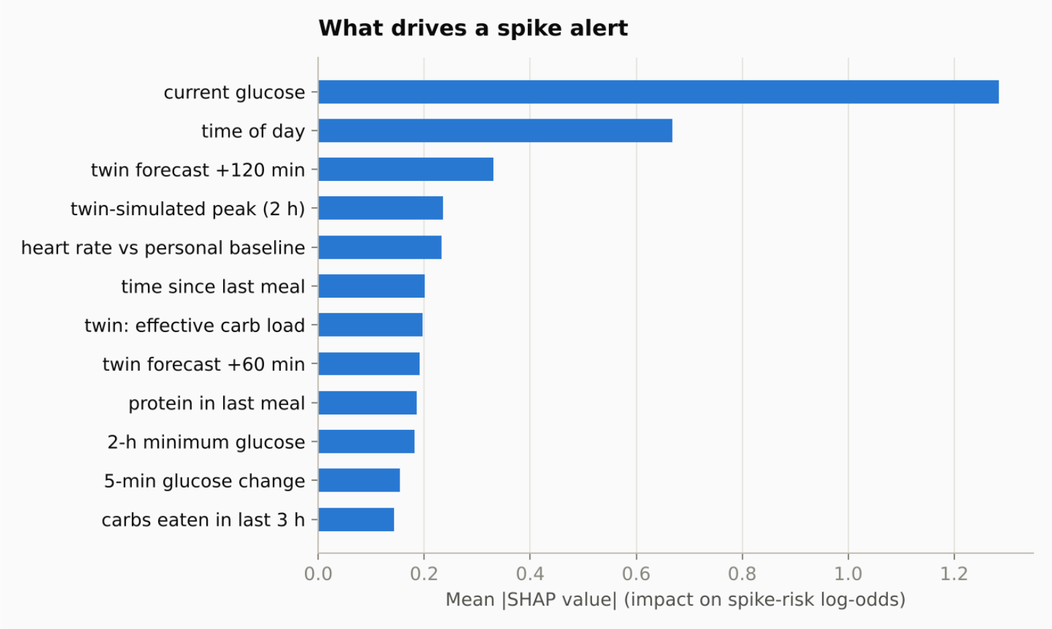
0.86
AUROC for a crossing above 180 mg/dL within 2 h

0.86
AUROC at the moment a meal is logged, before any rise

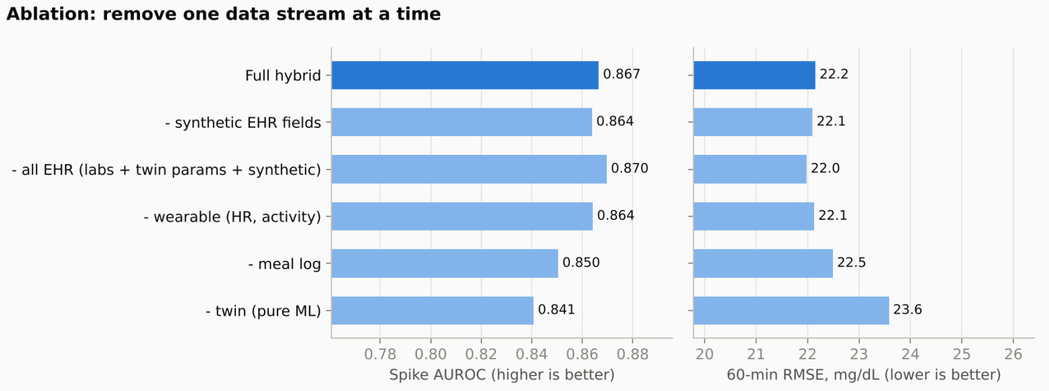
94%
of crossings caught, median 90 min ahead

2.1
false alarms per patient-day

Every alert explains itself, and every data stream earns its place



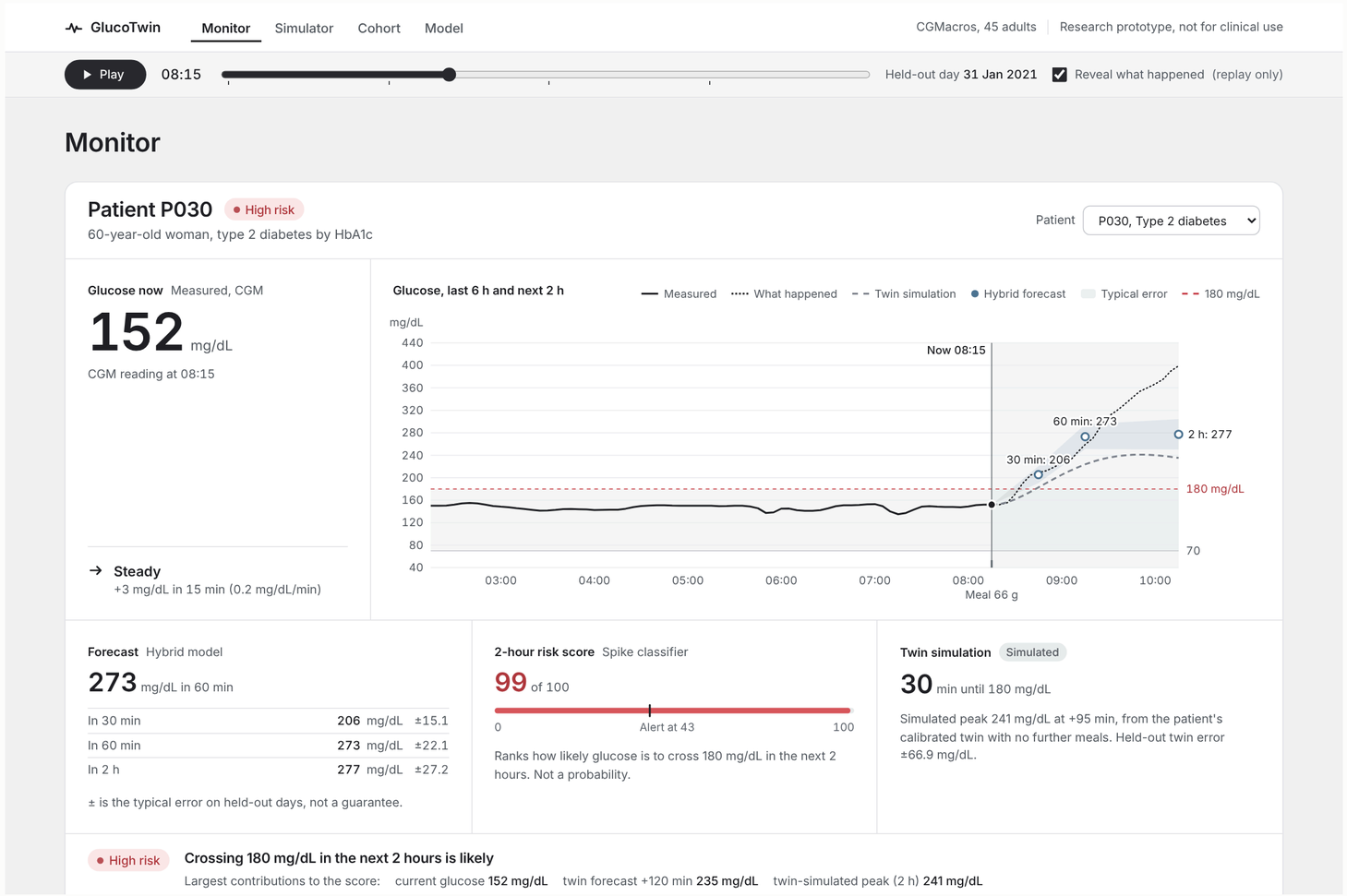
SHAP: what drives the spike-risk score



Remove one data stream at a time

The synthetic EHR fields were generated from the real labs, so they should add nothing. The ablation confirms it, and we report it instead of hiding it.

The doctor's console



Risk assessment with provenance, live twin and forecast, what-if with Indian meals, glucose profile, FHIR record

Why this matters for India

Personal diet advice

The what-if simulator speaks in Indian meals: idli-sambar, rajma chawal, dal-roti, biryani. Counselling becomes "for you, this", not generic rules.

Fits the national stack

Health records go in and out as FHIR R4, the format of the Ayushman Bharat Digital Mission, so a real ABHA-linked record could replace the synthetic one.

Cheap to run

The twin is a few equations. It runs in the doctor's browser with no server and no GPU, which matters for clinics outside metros.

Limitations and next steps

What this prototype is not

45 US adults over about 10 days: a proof of concept, not clinical validation.

Diagnoses, medication and genotype are synthetic. Plasma insulin was not measured.

Meal macros in the what-if library are approximate.

Not a medical device: decision support only.

Next

Validate on an Indian CGM cohort with Indian meals.

Add sleep and stress signals from the wearable.

Pilot with a diabetologist: do alerts change counselling?

Connect to real ABDM records with consent.

GlucoTwin

Code, results and the live dashboard are in the GitHub repository.

github.com/JiroRaiden/GlucoTwin_IIESTShibpur

Data: CGMacros (Gutierrez-Osuna, Kerr, Mortazavi, Das; PhysioNet 2025). Model: Bergman 1979, Dalla Man 2007. Clinical metrics: Clarke 1987, Battelino 2019 consensus on time in range.